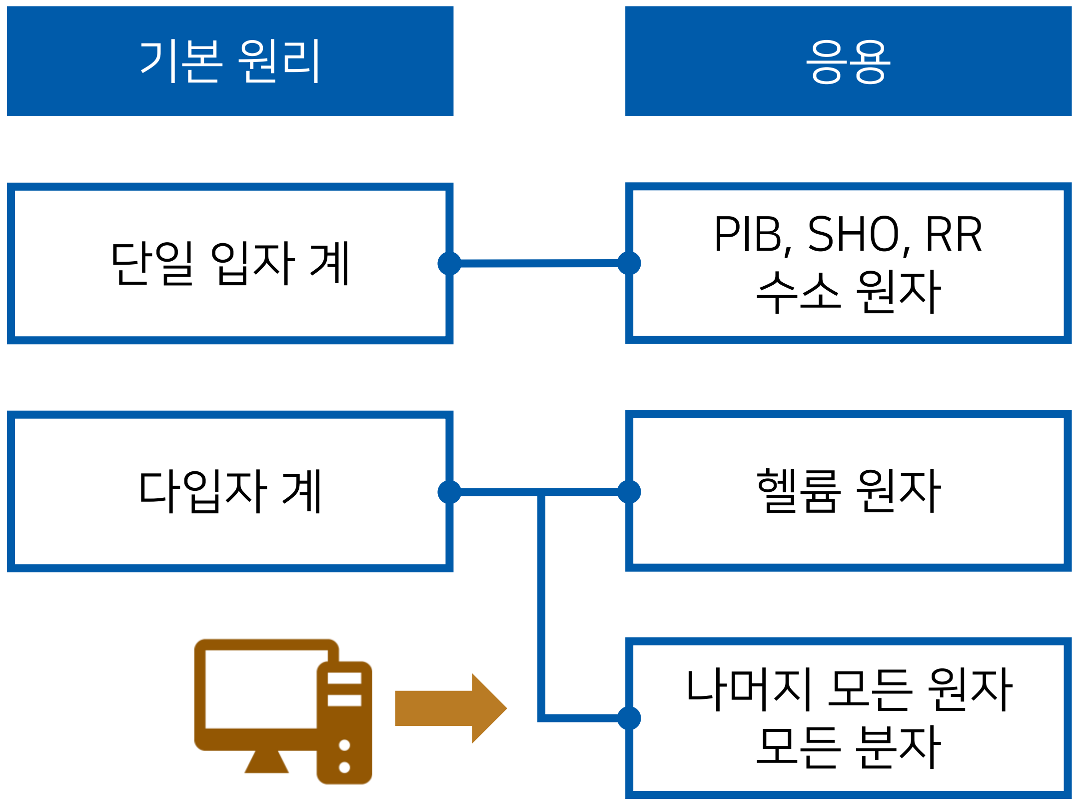
Physical Chemistry (II)

Lecture 17

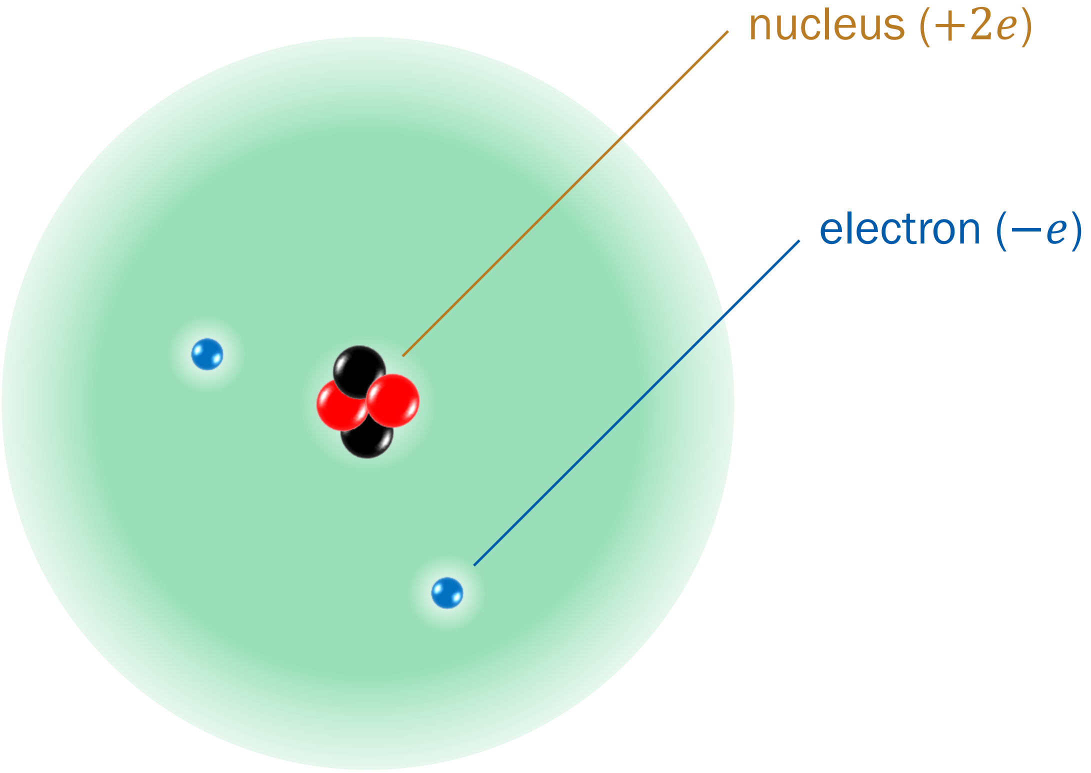
Hartree-Fock Method

Where Are We?



RS 7.1
M 7.9

Helium Atom



M 9.1

Ground-State Energy of He

Method	Energy (E_h)	Method	Energy (E_h)
Complete neglect of the interelectronic repulsion term			-4.0000
Perturbation calculations		*Variational calculations**	
First-order	-2.7500	1s orbitals (1)	-2.8477
Second-order	-2.9077	Eckart (2)	-2.8757
13^{th}-order	-2.90372433	Hartree-Fock (2)	-2.86168
		Hylleraas (10)	-2.90363
		Pekeris (1078)	-2.903724375
		Experimental value	-2.9033

* the number in parentheses indicates the number of parameters

Treating Helium

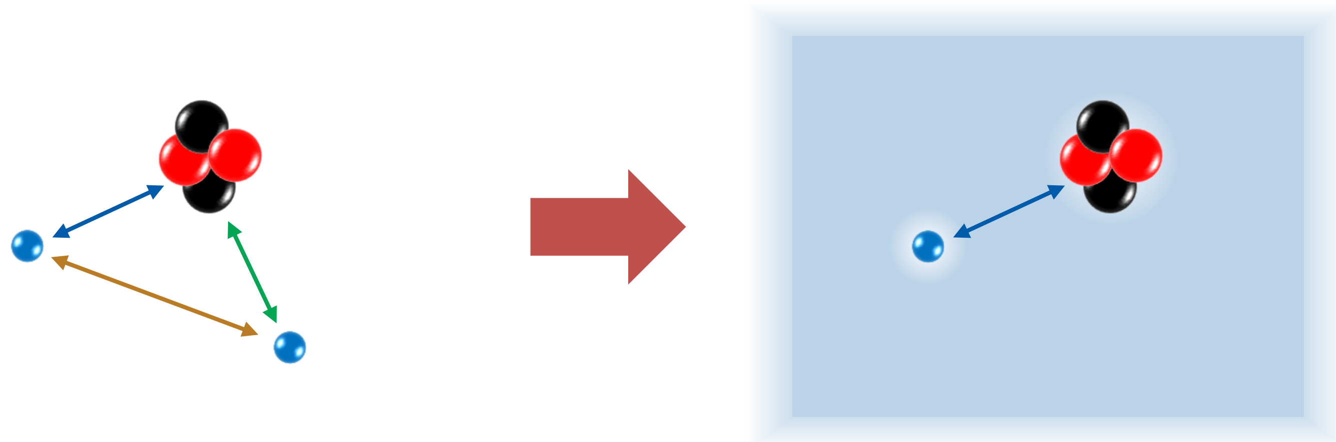
- Perturbation and variational approaches
  - can give very accurate energies
  - computationally difficult
  - not transferable to large atoms and molecules

- Orbital approximation
  - Each electron's spatial part is represented by an one-electron orbital $\psi_i(\vec{r})$
  - Common concept in chemistry
  - How to obtain "accurate" $\psi_i(\vec{r})$?

RS 9.3
M 9.3

Mean-Field Approach

$$\hat{H} = \underbrace{-\frac{\hbar^2}{2m_e}\nabla^2_{e1} - \frac{2e^2}{4\pi\varepsilon_0 r_1}}_{\hat{H}_{\mathrm{H1}}} \underbrace{-\frac{\hbar^2}{2m_e}\nabla^2_{e2} - \frac{2e^2}{4\pi\varepsilon_0 r_2}}_{\hat{H}_{\mathrm{H2}}} + \underbrace{\frac{e^2}{4\pi\varepsilon_0 |\vec{r}_1 - \vec{r}_2|}}_{\text{interelectronic repulsion}}$$



RS 9.3
M 9.3

Self-Consistent Field Method

- Trial function: $\phi(\vec{r}_1, \vec{r}_2) = \psi(\vec{r}_1)\psi'(\vec{r}_2)$
- Probability distribution of electron 2: $|\psi'(\vec{r}_2)|^2 \mathrm{d}\vec{r}_2$
- Potential energy of electron 1 due to electron 2:

$$V_1^{\mathrm{eff}}(\vec{r}_1) = \frac{e^2}{4\pi\varepsilon_0} \int \frac{1}{|\vec{r}_1 - \vec{r}_2|} |\psi'(\vec{r}_2)|^2 \, \mathrm{d}\vec{r}_2$$

RS 9.3
M 9.3

Self-Consistent Field Method

- Hamiltonian for electron 1:

$$\hat{H}_1^{\mathrm{eff}} = \hat{H}_{\mathrm{H1}} + V_1^{\mathrm{eff}}(\vec{r}_1) = -\frac{\hbar^2}{2m_e}\nabla_{e1}^2 - \frac{2e^2}{4\pi\varepsilon_0 r_1} + V_1^{\mathrm{eff}}(\vec{r}_1)$$

- Hartree-Fock (HF) equation:

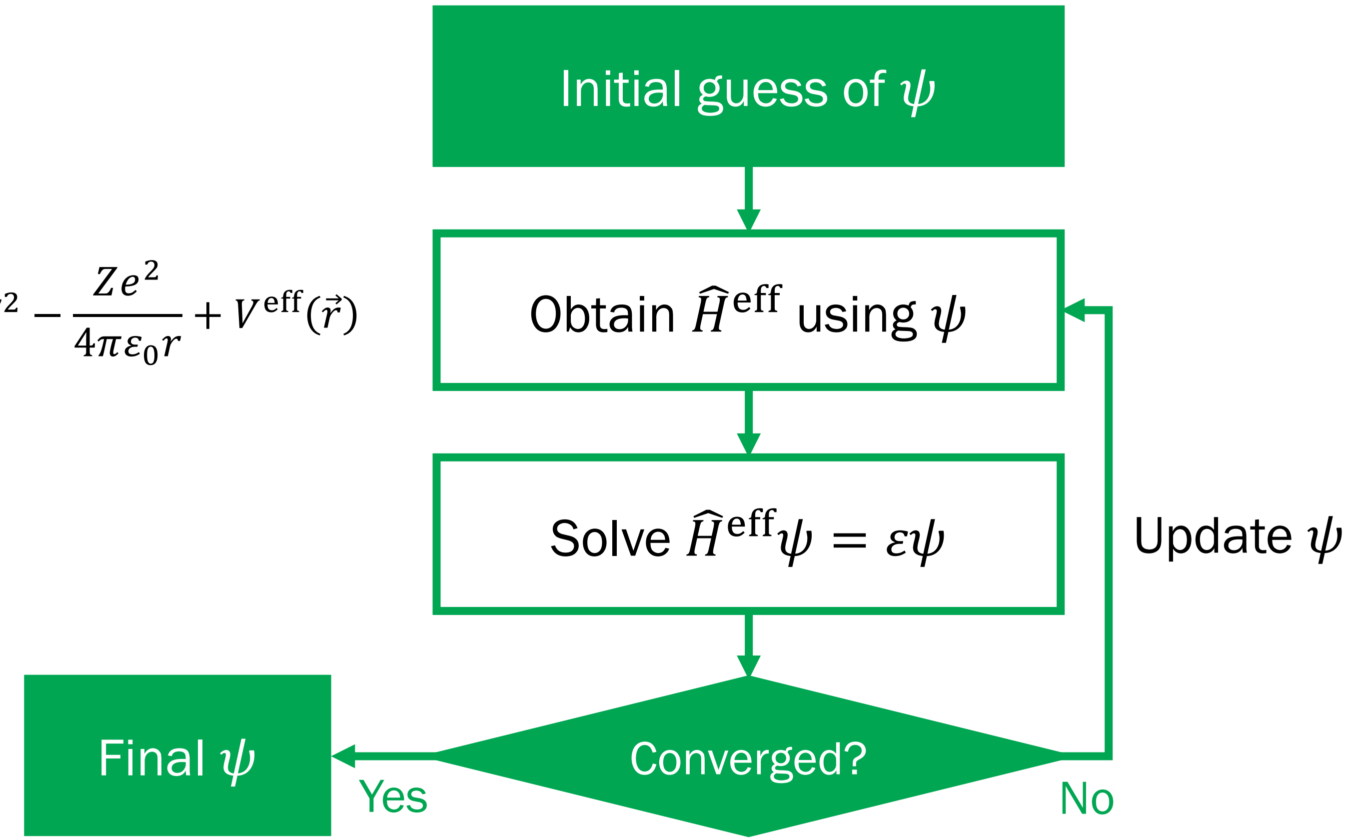
$$\hat{H}_1^{\mathrm{eff}}\psi(\vec{r}_1) = \varepsilon\psi(\vec{r}_1)$$

- Now solve the HF equation to obtain updated $\psi(\vec{r}_1)$, with which one can repeat the whole procedure

RS 9.3
M 9.3

Self-Consistent Field Method

$$\hat{H}^{\text{eff}} = -\frac{\hbar^2}{2m_e}\nabla^2 - \frac{Ze^2}{4\pi\varepsilon_0 r} + V^{\text{eff}}(\vec{r})$$



RS 9.3
M 9.3

One-Electron Wavefunctions

$$\hat{H}^{\text{eff}}\psi_i(\vec{r}) = \varepsilon_i\psi_i(\vec{r})$$

- Iteration gives a convergent solution: $\psi_i(\vec{r})$ and ε_i
- Still quantized (corresponding to H atom orbitals)
- Ground-state solution: $\psi_0(\vec{r})$ and ε_0
- The final two-electron solution:

$$\phi_{ij}(\vec{r}_1, \vec{r}_2) = \psi_i(\vec{r}_1)\psi_j(\vec{r}_2)$$

Example: Helium Atom

- Trial function: $\phi(\vec{r}_1, \vec{r}_2) = \psi(\vec{r}_1)\psi(\vec{r}_2)$

$$\psi(\vec{r}) = c_1 \left(\frac{\zeta_1^3}{\pi}\right)^{1/2} e^{-\zeta_1 r} + c_2 \left(\frac{\zeta_2^3}{\pi}\right)^{1/2} e^{-\zeta_2 r}$$

with fixed ζ_1 = 1.45363 and ζ_2 = 2.91093

Iteration	c_1	c_2	E / hartrees
1	0.500	0.500	-2.394
2	0.884	0.135	-2.857
3	0.836	0.190	-2.862
4	0.845	0.179	-2.862
5	0.844	0.181	-2.862
⋮			
10	0.844	0.181	-2.862

RS 8.3
M 9.4

Electronic Wavefunction

- Must contain spin wavefunction

$$\phi_{ij}(\vec{r}_1,\vec{r}_2;\sigma_1,\sigma_2) = \psi_i(\vec{r}_1)\psi_j(\vec{r}_2)|\sigma_1\rangle_1|\sigma_2\rangle_2$$

- Must be antisymmetric

$$\phi_{ij}(\vec{r}_1,\vec{r}_2;\sigma_1,\sigma_2) = -\phi_{ij}(\vec{r}_2,\vec{r}_1;\sigma_2,\sigma_1)$$

- Pauli exclusion principle (another version): two identical fermions cannot be detected <u>at the same state</u>

RS 8.3
M 9.4

Ground-State Wavefunction

$$\phi_{ij}(\vec{r}_1, \vec{r}_2; \sigma_1, \sigma_2) = \psi_i(\vec{r}_1)\psi_j(\vec{r}_2)|\sigma_1\rangle_1|\sigma_2\rangle_2$$

- The system energy only depends on the spatial part (unless a magnetic field is applied)
- The lowest energy is obtained when

$$\psi_i(\vec{r}_1) = \psi_0(\vec{r}_1)$$

$$\psi_j(\vec{r}_2) = \psi_0(\vec{r}_2)$$

RS 8.3
M 9.4

Ground-State Wavefunction

$$\phi_{00}(\vec{r}_1, \vec{r}_2; \sigma_1, \sigma_2) = \psi_0(\vec{r}_1)\psi_0(\vec{r}_2)|\sigma_1\rangle_1|\sigma_2\rangle_2$$

- The spatial wavefunction is symmetric
- The spin wavefunction must be antisymmetric: singlet

$$\frac{1}{\sqrt{2}}(|\alpha\rangle_1|\beta\rangle_2 - |\beta\rangle_1|\alpha\rangle_2)$$

- The total wavefunction

$$\phi_{00}(\vec{r}_1, \vec{r}_2) = \frac{1}{\sqrt{2}}\psi_0(\vec{r}_1)\psi_0(\vec{r}_2)(|\alpha\rangle_1|\beta\rangle_2 - |\beta\rangle_1|\alpha\rangle_2)$$

RS 9.3
M 9.3

Ground-State Energy

$$\hat{H}_0 = \hat{H}_{\mathrm{H1}} + \hat{H}_{\mathrm{H2}}, \qquad \hat{H}' = \frac{e^2}{4\pi\varepsilon_0 |\vec{r}_1 - \vec{r}_2|}$$

- Zeroth-order energy

$$\hat{H}_0 \psi_0(\vec{r}_1)\psi_0(\vec{r}_2) = (E_0 + E_0)\psi_0(\vec{r}_1)\psi_0(\vec{r}_2)$$

- First-order energy

$$E^{(1)} = \int \psi_0^*(\vec{r}_1)\psi_0^*(\vec{r}_2)\hat{H}'\psi_0(\vec{r}_1)\psi_0(\vec{r}_2)\,\mathrm{d}\vec{r}_1\mathrm{d}\vec{r}_2 = J_{00}$$

$$\rightarrow E = 2E_0 + J_{00}$$

RS 9.3
M 9.3

Orbital Energies

$$\hat{H}_1^{\text{eff}}\psi_i(\vec{r}) = \varepsilon_i\psi_i(\vec{r})$$

$$\hat{H}_1^{\text{eff}} = \hat{H}_{\text{H1}} + V_1^{\text{eff}}(\vec{r}_1)$$

- Zeroth-order energy

$$\hat{H}_{\text{H1}}\psi_0(\vec{r}_1) = E_0\psi_0(\vec{r}_1)$$

- First-order energy

$$E^{(1)} = \int \psi_0^*(\vec{r}_1)V_1^{\text{eff}}(\vec{r}_1)\psi_0(\vec{r}_1)\,\mathrm{d}\vec{r}_1$$

RS 9.3
M 9.3

First-Order Energy

$$E^{(1)} = \int \psi_0^*(\vec{r}_1) V_1^{\text{eff}}(\vec{r}_1) \psi_0(\vec{r}_1) \, \mathrm{d}\vec{r}_1$$

$$V_1^{\text{eff}}(\vec{r}_1) = \frac{e^2}{4\pi\varepsilon_0} \int \frac{1}{|\vec{r}_1 - \vec{r}_2|} |\psi_0(\vec{r}_2)|^2 \, \mathrm{d}\vec{r}_2$$

Substitution leads to

$$E^{(1)} = \frac{e^2}{4\pi\varepsilon_0} \int \psi_0^*(\vec{r}_1) \psi_0^*(\vec{r}_2) \frac{1}{|\vec{r}_1 - \vec{r}_2|} \psi_0(\vec{r}_1) \psi_0(\vec{r}_2) \, \mathrm{d}\vec{r}_1 \mathrm{d}\vec{r}_2 = J_{00}$$

$$\rightarrow \varepsilon_0 = E_0 + J_{00}$$

RS 9.3
M 9.3

Orbital Energies

- Ground-state energy $E = 2E_0 + J_{00}$
- Orbital energy $\varepsilon_0 = E_0 + J_{00}$

$$E \neq 2\varepsilon_0$$

(total energy of He $\neq$ sum of orbital energies)

RS 9.3
M 9.3

Koopman's Approximation

- Ground-state energy $E = 2E_0 + J_{00}$
- Orbital energy $\varepsilon_0 = E_0 + J_{00}$

$$E = E_0 + \varepsilon_0 \longrightarrow \varepsilon_0 = E - E_0$$

- E: energy of He atom
- E_0: energy of He^+ ion (hydrogen-like atom with Z = 2)
- $E_0 - E$: ionization energy (IE) of He atom

$$\mathrm{IE} \approx -\varepsilon_0$$

RS 9.2
M 9.5

Case of Three Electrons (Li)

$$\phi_{ijk}(1,2,3) = \psi_i(\vec{r}_1)\psi_j(\vec{r}_2)\psi_k(\vec{r}_3)|\sigma_1\rangle_1|\sigma_2\rangle_2|\sigma_3\rangle_3$$

- Separation of spatial and spin variables: one cannot find an antisymmetric spin wavefunction

- Need a systematic way to find a total antisymmetric wavefunction

RS 9.2
M 9.5

Slater Determinant

- Two-electron system: for two quantum states i and j,

$$\phi_{ij}(1,2) = \frac{1}{\sqrt{2}} \begin{vmatrix} u_i(1) & u_j(1) \\ u_i(2) & u_j(2) \end{vmatrix}$$

Particle 1 in state j (including both spatial and spin parts)

- N-electron system: for N quantum states $a, b, \cdots, n$,

$$\phi_{ab\cdots n}(1,2,\cdots,N) = \frac{1}{\sqrt{N!}} \begin{vmatrix} u_a(1) & u_b(1) & \cdots & u_n(1) \\ u_a(2) & u_b(2) & \cdots & u_n(2) \\ \vdots & \vdots & \ddots & \vdots \\ u_a(N) & u_b(N) & \cdots & u_n(N) \end{vmatrix}$$

Review

Determinant of a Square Matrix

$$\det \mathbf{A} = \begin{vmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{vmatrix}$$

- 2 by 2 determinant

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

Expanding the Determinant

1. Pick a row or a column (with many zeros)
2. Calculate each element × sign × minor
   - Minor: determinant after deleting the row and the column containing the element
   - Sign: counting the number of steps of one row or one column required to get from the upper left element
     → even number: (+), odd number: (-)
3. Sum them up across the row or the column
4. Repeat this up to a sum of 2 by 2 determinants

Example

Expand the 3 by 3 determinant of a general matrix $\mathbf{A}$.

$$\det \mathbf{A} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

Pick the first row:

$$\det \mathbf{A} = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

Now we can apply the definition of the 2 by 2 determinant:

$$\det \mathbf{A} = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31})$$

RS 9.2
M 9.5

Property 1 of Determinant

$$\phi_{ab\cdots n}(1,2,\cdots,N)=\frac{1}{\sqrt{N!}}\begin{vmatrix} u_a(1) & u_b(1) & \cdots & u_n(1) \\ u_a(2) & u_b(2) & \cdots & u_n(2) \\ \vdots & \vdots & \ddots & \vdots \\ u_a(N) & u_b(N) & \cdots & u_n(N) \end{vmatrix}$$

- If two rows or two columns are interchanged, the determinant = (-1) × (original determinant).

  → Antisymmetric property

RS 9.2
M 9.5

Property 2 of Determinant

$$\phi_{ab\cdots n}(1,2,\cdots,N) = \frac{1}{\sqrt{N!}} \begin{vmatrix} u_a(1) & u_b(1) & \cdots & u_n(1) \\ u_a(2) & u_b(2) & \cdots & u_n(2) \\ \vdots & \vdots & \ddots & \vdots \\ u_a(N) & u_b(N) & \cdots & u_n(N) \end{vmatrix}$$

- If two rows or two columns are identical, the determinant has value zero.

  $\longrightarrow$ Pauli exclusion principle

Lithium Again

- One-electron states: $\psi_0|\alpha\rangle, \psi_0|\beta\rangle, \psi_1|\alpha\rangle$
- Slater determinant

$$\phi(1,2,3) = \frac{1}{\sqrt{3!}}\begin{vmatrix} \psi_0(\vec{r}_1)|\alpha\rangle_1 & \psi_0(\vec{r}_1)|\beta\rangle_1 & \psi_1(\vec{r}_1)|\alpha\rangle_1 \\ \psi_0(\vec{r}_2)|\alpha\rangle_2 & \psi_0(\vec{r}_2)|\beta\rangle_2 & \psi_1(\vec{r}_2)|\alpha\rangle_2 \\ \psi_0(\vec{r}_3)|\alpha\rangle_3 & \psi_0(\vec{r}_3)|\beta\rangle_3 & \psi_1(\vec{r}_3)|\alpha\rangle_3 \end{vmatrix}$$

⟶ always antisymmetric!

Shorthand Notations

- α and β by unbarred and barred spatial wavefunctions

$$\phi(1,2,3) = \frac{1}{\sqrt{3!}}\begin{vmatrix} \psi_0(\vec{r}_1)|\alpha\rangle_1 & \psi_0(\vec{r}_1)|\beta\rangle_1 & \psi_1(\vec{r}_1)|\alpha\rangle_1 \\ \psi_0(\vec{r}_2)|\alpha\rangle_2 & \psi_0(\vec{r}_2)|\beta\rangle_2 & \psi_1(\vec{r}_2)|\alpha\rangle_2 \\ \psi_0(\vec{r}_3)|\alpha\rangle_3 & \psi_0(\vec{r}_3)|\beta\rangle_3 & \psi_1(\vec{r}_3)|\alpha\rangle_3 \end{vmatrix}$$

$$= \frac{1}{\sqrt{3!}}\begin{vmatrix} \psi_0(1) & \overline{\psi_0(1)} & \psi_1(1) \\ \psi_0(2) & \overline{\psi_0(2)} & \psi_1(2) \\ \psi_0(3) & \overline{\psi_0(3)} & \psi_1(3) \end{vmatrix}$$

$$= |\psi_0 \quad \overline{\psi_0} \quad \psi_1|$$

RS 9.4
M 9.6

Hartree-Fock Method

- The self-consistent field (SCF) method
- The atomic wavefunction = Slater determinant

- Electrons interact only through an average potential

  $\longrightarrow$ They are "uncorrelated"

- (Electron) correlation energy E_{corr}

$$E_{\mathrm{corr}} = E_{\mathrm{exact}} - E_{\mathrm{HF}}$$

RS 9.4
M 9.6

N-Electron Hamiltonian

For an N-electron atom,

$$\hat{H} = \sum_{j=1}^{N}\left(-\frac{\hbar^2}{2m_e}\nabla_j^2 - \frac{Ze^2}{4\pi\varepsilon_0 r_j}\right) + \sum_{j=1}^{N}\sum_{i<j}\frac{e^2}{4\pi\varepsilon_0\left|\vec{r}_i - \vec{r}_j\right|}$$

or

$$\hat{H} = \sum_{j=1}^{N}\hat{H}_j + \sum_{j=1}^{N}\sum_{i<j}\frac{e^2}{4\pi\varepsilon_0\left|\vec{r}_i - \vec{r}_j\right|}$$

RS 9.4
M 9.6

Ground-State Wavefunction

For a closed configuration, by letting $N = 2M$,

the ground-state wavefunction is

$$\phi(1, 2, \cdots, 2M) = |\psi_0 \quad \overline{\psi_0} \quad \psi_1 \quad \overline{\psi_1} \quad \cdots \quad \psi_{M-1} \quad \overline{\psi_{M-1}}|$$

$$\hat{H} = \underbrace{\sum_{j=1}^{2M} \hat{H}_j}_{\hat{H}_0} + \underbrace{\sum_{j=1}^{2M} \sum_{i<j} \frac{e^2}{4\pi\varepsilon_0 |\vec{r}_i - \vec{r}_j|}}_{\hat{H}'}$$

Ground-State Energy

- Zeroth-order term

$$\hat{H}_0\phi(1,2,\cdots,2M) = \left(\sum_{j=1}^{M} 2E_j\right)\phi(1,2,\cdots,2M)$$

- First-order term

$$E^{(1)} = \int \phi^*(1,2,\cdots,2M)\hat{H}'\phi(1,2,\cdots,2M)\,\mathrm{d}\vec{r}_1\mathrm{d}\vec{r}_2\cdots\mathrm{d}\vec{r}_{2M}$$

First-Order Term

$$E^{(1)} = \int \phi^*(1,2,\cdots,2M)\widehat{H}'\phi(1,2,\cdots,2M)\,\mathrm{d}\vec{r}_1\mathrm{d}\vec{r}_2\cdots\mathrm{d}\vec{r}_{2M}$$

$$J_{ij} = \int \psi_i^*(\vec{p})\psi_j^*(\vec{q})\frac{e^2}{4\pi\varepsilon_0|\vec{p}-\vec{q}|}\psi_i(\vec{p})\psi_j(\vec{q})\,\mathrm{d}\vec{p}\mathrm{d}\vec{q}$$

$$K_{ij} = \int \psi_i^*(\vec{p})\psi_j^*(\vec{q})\frac{e^2}{4\pi\varepsilon_0|\vec{p}-\vec{q}|}\psi_i(\vec{q})\psi_j(\vec{p})\,\mathrm{d}\vec{p}\mathrm{d}\vec{q}$$

$$E^{(1)} = \sum_{i=1}^{M}\sum_{j=1}^{M}\left(2J_{ij}-K_{ij}\right)$$

Ground-State Energy

$$E = \sum_{j=1}^{M} 2E_j + \sum_{i=1}^{M}\sum_{j=1}^{M}\left(2J_{ij} - K_{ij}\right)$$

Similarly, the orbital energy can be shown to be

$$\varepsilon_i = E_i + \sum_{j=1}^{M}\left(2J_{ij} - K_{ij}\right)$$

$$\therefore\ E = \sum_{i=1}^{M}\left(E_i + \varepsilon_i\right)$$

Ionization Energy

If an electron in orbital k is dissociated, the energy of the cation is (roughly)

$$E^{+} = \sum_{j=1}^{M} 2E_j - E_k + \sum_{i=1}^{M}\sum_{j=1}^{M}\left(2J_{ij} - K_{ij}\right) - \sum_{i=1}^{M}\left(2J_{ik} - K_{ik}\right)$$

The ionization energy is

$$\mathrm{IE} = E^{+} - E = -E_k - \sum_{i=1}^{M}\left(2J_{ik} - K_{ik}\right) = -\varepsilon_k$$

RS 9.5
M 9.7

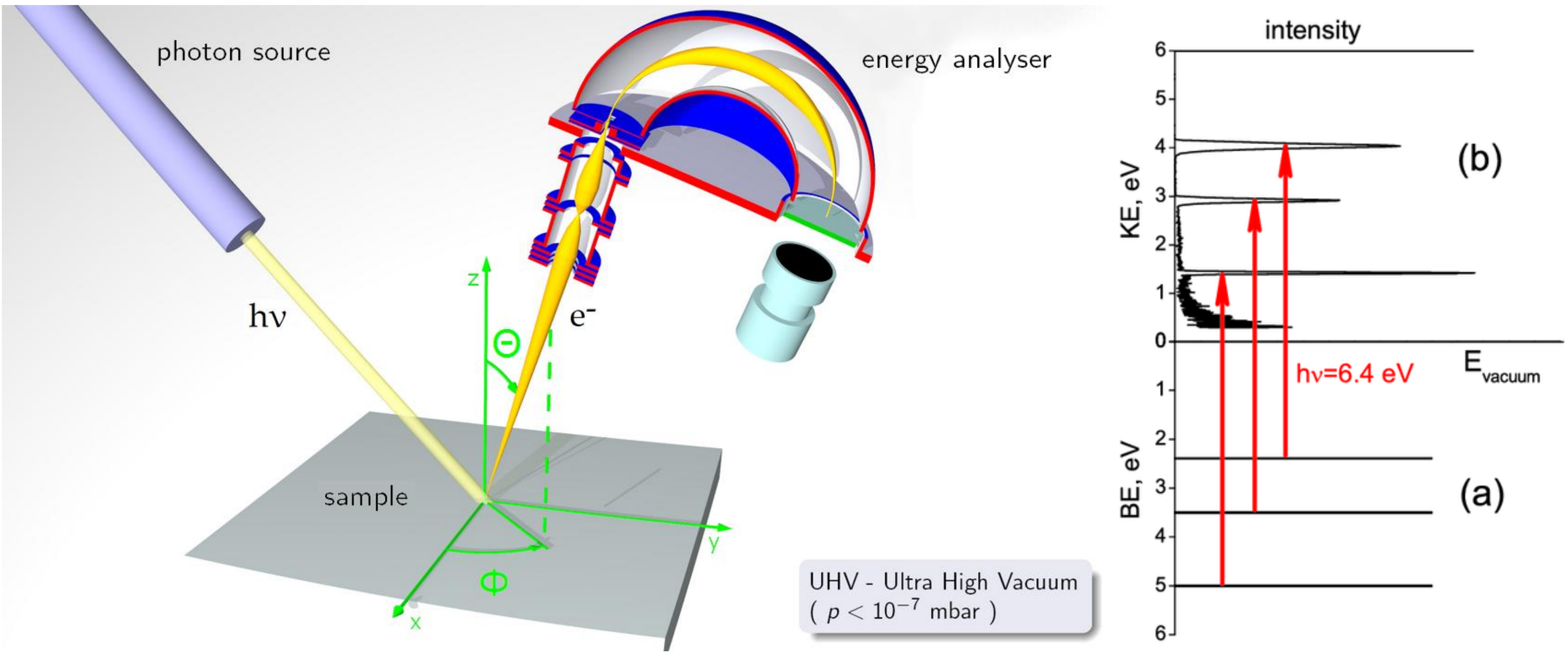
Koopman's Approximation

$$\text{IE} \approx -\varepsilon_k$$

- Ionization energy gives information about orbital energies
- Good approximation for closed-shell systems

Not in text

Photoelectron Spectroscopy



(Image: https://en.wikipedia.org/wiki/File:ARPESgeneral.png
https://www.researchgate.net/figure/Principle-of-photoelectron-spectroscopy-Shown-as-an-example-is-the-PES-of-gold-anions_fig9_29757299)

Ionization Energies

Electron removed	Resulting orbital occupancy	Ionization energies / MJ·mol^{-1}		
		Koopman's approximation	HF calculation	Experimental
Neon				
1*s*	$1s^1 2s^2 2p^6$	86.0	83.80	83.96
2*s*	$1s^2 2s^1 2p^6$	5.07	4.76	4.68
2*p*	$1s^2 2s^2 2p^5$	2.23	1.92	2.08
Argon				
1*s*	$1s^1 2s^2 2p^6 3s^2 3p^6$	311.4	308.25	309.32
2*s*	$1s^2 2s^1 2p^6 3s^2 3p^6$	32.35	31.33	
2*p*	$1s^2 2s^2 2p^5 3s^2 3p^6$	25.13	24.01	23.97
3*s*	$1s^2 2s^2 2p^6 3s^1 3p^6$	3.35	3.20	2.82
3*p*	$1s^2 2s^2 2p^6 3s^2 3p^5$	1.55	1.43	1.52